

Kshitij Rastogi

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Professional Summary

Final-year B.Tech Information Technology student with hands-on experience building AI-powered, full-stack applications using Large Language Models, Retrieval-Augmented Generation (RAG), LangChain, and LangGraph. Skilled in developing intelligent healthcare systems, multi-agent AI workflows, REST APIs, and scalable web applications using JavaScript, TypeScript, Python, React.js and Node.js. Proven track record of delivering production-ready software and competitive results in AI-driven hackathons.

Technical Skills

Languages: JavaScript, TypeScript, Python

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Three.js, SCSS

Backend: Node.js, Express.js, REST APIs

AI & Generative AI: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, OpenAI API, Gemini API, Claude API, Mistral, AutoGen

Databases: MongoDB, Redis, Firebase

Tools & DevOps: Docker, Kubernetes, Skkafold, Git, GitHub, JWT Authentication, Google OAuth, Node Mailer

Experience

Student Research Associate, Virtual Labs Internship

Feb2026–Apr2026

Indian Institute of Technology (IIT) Kanpur

- Contributed to development of interactive virtual laboratory modules for educational use, building web-based simulations for research-oriented software projects.
- Collaborated with a research team to design, implement, and test simulation modules, improving problem-solving, software development, and teamwork skills.

Projects

PulseTalk – AI-Powered Healthcare Assistant

LiveDemo: pulsetalk.vercel.app

- Built a full-stack AI healthcare platform using TypeScript, Express.js, MongoDB, Gemini API, JWT Authentication, and Google OAuth and secure authentication.
- Implemented AI chat, voice assistant, Retrieval-Augmented Generation (RAG), appointment booking, chat history,
- Developed disease prediction through image analysis with automated medicine recommendations.
- Designed scalable REST APIs and integrated intelligent, end-to-end healthcare workflows.

LLM Evaluation Framework – Multi-Agent AI System

LiveDemo: aibattlearena

- Developed a multi-agent AI workflow using LangChain and LangGraph to route each user prompt to two independent LLMs for parallel response generation.
- Implemented an LLM-as-a-Judge pipeline where a third model evaluated and scored both responses for quality and accuracy
- Designed a modular architecture supporting additional models and evaluation strategies.

Virtual Laboratory Platform

LiveDemo: simulation.vlabsiitk

- Developed interactive educational simulations during the IIT Kanpur Virtual Lab internship.
- Built reusable web components and contributed to research-focused learning experiences.

Achievements

- Top 14 Finalist among 500 teams (3200+ registrations) at GDG Delhi HackFest for proposing an AI-powered healthcare diagnosis platform.
- Top 10 Finalist among 250 teams (1500+ registrations) at GDG Noida Hackathon.
- Solved 150+ Data Structures and Algorithms problems across competitive coding platforms.

Education

Bachelor of Technology in Information Technology

2023–2027

CGPA: 8.0/10